ECE 6950, AQOPT, Fall 2024
Prof. Karan Mehta

Problem Set 4: Quantized fields, JC model

Assigned: Tuesday, Oct. 1
Due: Thursday, Oct. 10

Question 1. In this exercise, we consider the Q-function introduced in lecture, and defined for a pure state $|\Psi\rangle$ as

$$Q_{|\Psi\rangle}(\beta) \equiv \frac{1}{\pi} |\langle\beta|\Psi\rangle|^2 , \tag{1}$$

and look at some examples.

a) Calculate the inner product between two different coherent states, i.e. $\langle\alpha_1|\alpha_2\rangle$; are coherent states with different eigenvalues orthogonal?
Recalling the expansion of the coherent states in terms of number states

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \tag{2}$$

we can write

$$\begin{aligned} \langle\alpha_1|\alpha_2\rangle &= e^{-|\alpha_1|^2/2} e^{-|\alpha_2|^2/2} \sum_m \sum_n \frac{(\alpha_1^*)^m}{\sqrt{m!}} \langle m| \frac{\alpha_2^n}{\sqrt{n!}} |n\rangle && (3) \\ &= e^{-\left(|\alpha_1|^2+|\alpha_2|^2\right)/2} \sum_n \frac{(\alpha_1^*\alpha_2)^n}{n!} && (4) \\ &= e^{-\left(|\alpha_1|^2+|\alpha_2|^2\right)/2} e^{\alpha_1^*\alpha_2} && (5) \end{aligned}$$

Where we have used the orthonormality of the number states, $\langle m|n\rangle = \delta_{mn}$, and from which we see that the coherent states are normalized ($\langle\alpha|\alpha\rangle = 1$) but not orthogonal to each other.

b) The definition of the Q-function as a projection onto coherent states makes sense because the coherent states span the full harmonic oscillator Hilbert space; the normalization factor $\frac{1}{\pi}$ comes from the fact that the coherent states are in fact overcomplete. Recalling that the number states are orthonormal and form a complete set, that is

$$\sum_{n=0}^{\infty} |n\rangle\langle n| = \hat{I} \tag{6}$$

with $\hat{I}$ the identity operator, show that

$$\int |\alpha\rangle\langle\alpha| d^2\alpha = \pi\hat{I}, \tag{7}$$

which expresses the coherent states' overcompleteness. Here $d^2\alpha$ represents 2D integration over both the real and imaginary parts of α in the complex plane.

Hint: to do this you can start by considering the matrix elements of this operator in the number state basis, i.e.

$$\langle n| \left(\int |\alpha\rangle\langle\alpha| d^2\alpha \right) |m\rangle, \tag{8}$$

and you may use the integral formula for the factorial:

$$n! = \int_0^\infty x^n e^{-x} dx \tag{9}$$

We start by considering the matrix elements

$$\langle n| \left(\int |\alpha\rangle\langle\alpha| d^2\alpha \right) |m\rangle = \int (\langle n|\alpha\rangle)(\langle\alpha|m\rangle) d^2\alpha \tag{10}$$

$$= \int \frac{(\alpha^*)^n \alpha^m}{\sqrt{n!m!}} e^{-|\alpha|^2} d^2\alpha \tag{11}$$

$$= \int_0^\infty z dz \int_0^{2\pi} d\varphi \frac{z^{n+m} e^{i(m-n)\varphi}}{\sqrt{n!m!}} e^{-z^2} \tag{12}$$

where in the last step to evaluate the integral over the complex plane, we have defined $\alpha = ze^{i\varphi}$. When $m \neq n$, it is easy to see that this integral evaluates to zero due to the exponential term oscillating with φ. For $m = n$, we have

$$\langle n| \left(\int |\alpha\rangle\langle\alpha| d^2\alpha \right) |n\rangle = 2\pi \int_0^\infty \frac{z^{2n+1}}{n!} e^{-z^2} dz \tag{13}$$

$$= \pi \int_0^\infty \frac{y^n}{n!} e^{-y} dy \tag{14}$$

$$= \pi \tag{15}$$

where the second-to-last step used the change of variable $y = z^2$, and the last step used the integral formula for the factorial. Since all off-diagonal elements evaluate to 0 and diagonal elements evaluate to π, we have shown that

$$\int |\alpha\rangle\langle\alpha| d^2\alpha = \pi \hat{I}. \tag{16}$$

c) Use this result to easily show that the Q-function is then a normalized distribution function, i.e.

$$\int Q_{|\Psi\rangle}(\beta) d^2\beta = 1 \tag{17}$$

for any normalized wavefunction $|\Psi\rangle$.
We expand

$$\int Q_{|\Psi\rangle}(\beta) d^2\beta = \frac{1}{\pi} \int \langle\Psi|\beta\rangle\langle\beta|\Psi\rangle d^2\beta \tag{18}$$

$$= \frac{1}{\pi} \langle\Psi| \left(\int |\beta\rangle\langle\beta| d^2\beta \right) |\Psi\rangle \tag{19}$$

$$= \frac{1}{\pi} \pi \langle\Psi|\Psi\rangle = 1. \tag{20}$$

Calculate, plot an illustrative example or two, and describe the features of the Q-function for the Fock states: $Q_{|n\rangle}(\beta)$, for the coherent states: $Q_{|\alpha\rangle}(\beta)$.
Using the result of part (a) we see that

$$Q_{|\alpha\rangle}(\beta) = \frac{1}{\pi} |\langle\alpha|\beta\rangle|^2 = \frac{1}{\pi} e^{-\left(|\alpha|^2 + |\beta|^2\right)} e^{2\mathrm{Re}(\alpha^*\beta)} \tag{21}$$

$$= \frac{1}{\pi} e^{-|\alpha-\beta|^2} \tag{22}$$

For a number state, the Q-function evaluates simply to

$$Q_{|n\rangle}(\beta) = \frac{1}{\pi} e^{-|\beta|^2} \frac{|\beta|^{2n}}{n!}. \tag{23}$$

These functions are plotted below. Fock states with nonzero n depend only on $|\beta|$ and hence appear as rings, with a radius $\sqrt{n}$ (it is easy to verify $Q_{|n\rangle}(\beta)$ is maximized when $|\beta| = \sqrt{n}$). The dependence only on $|\beta|$ reflects number states' undefined phase. A coherent state with $\alpha = 0$ is simply the Fock state with $n = 0$ and so isn't plotted; a coherent state with nonzero α appears simply as this function displaced in the complex plane by α.

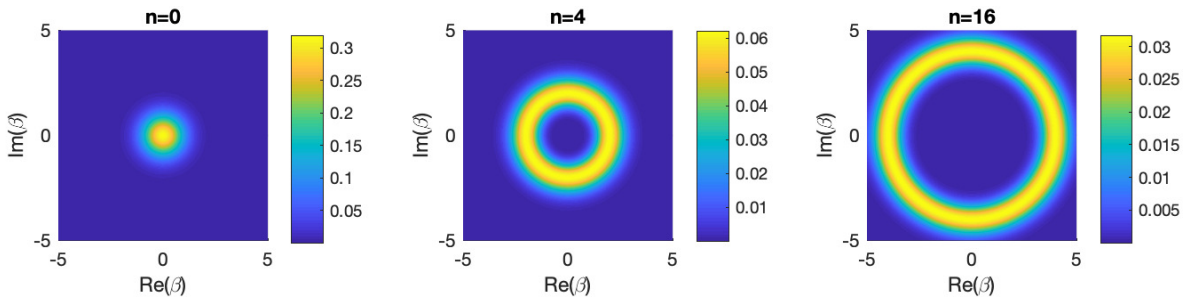


FIGURE 1. Q-functions for Fock states with values of n labeled

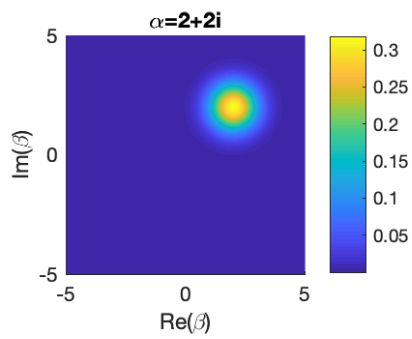


FIGURE 2. Q-function for a coherent state with $\alpha = 2 + 2i$

Question 2. Limits in the Jaynes-Cummings model

In class we looked at the problem of collapse and revival of oscillations in a JC model with cavity mode starting in a coherent state.

a) Calculate $\langle \hat{n} \rangle$ and $\langle \hat{n}^2 \rangle$, and σ_n for a coherent state $|\alpha\rangle$.
$\langle \hat{n} \rangle = |\alpha|^2$, $\langle \hat{n}^2 \rangle = |\alpha|^2 (|\alpha|^2 + 1)$, and so $\sigma_n = |\alpha| = \sqrt{\langle \hat{n} \rangle}$. For large $\langle \hat{n} \rangle$, the standard deviation relative to the mean becomes very small.

b) Consider the model we developed in class, i.e. a starting state $\psi(0) = |e\rangle \otimes |\alpha\rangle$, with each $|e, n\rangle$ component resulting in Rabi oscillations as:

$$|\psi_{e,n}(t)\rangle = \cos(\omega_{n+1} t)|e, n\rangle - i \sin(\omega_{n+1} t)|g, n+1\rangle, \tag{24}$$

with $\omega_{n+1} = g_0 \sqrt{n+1}$ and with the starting $|n\rangle$ states distributed as a coherent state $|\alpha\rangle$. Consider the limit where α is very large, and to first order given the result for σ_n above, approximate $\sqrt{n+1} \approx \sqrt{\langle n \rangle}$. Show that the time evolution is then consistent with the semiclassical expectation.

For large $\langle n \rangle$ and with $c_n = e^{-|\alpha|^2/2} \frac{\alpha^n}{\sqrt{n!}}$ we have

$$\begin{aligned} |\psi(t)\rangle &= \sum_n c_n \left(\cos(\omega_{n+1} t)|e, n\rangle - i \sin(\omega_{n+1} t)|g, n+1\rangle\right) && (25) \\ &\approx \sum_n c_n \left(\cos(g_0 \sqrt{\langle n \rangle} t)|e, n\rangle - i \sin(g_0 \sqrt{\langle n \rangle} t)|g, n\rangle\right) && (26) \\ &= \left(\cos(g_0 \sqrt{\langle n \rangle} t)|e\rangle - i \sin(g_0 \sqrt{\langle n \rangle} t)|g\rangle\right) \otimes \sum_n c_n |n\rangle && (27) \end{aligned}$$

corresponding to Rabi oscillations between $|e\rangle$ and $|g\rangle$ at the a Rabi frequency proportional to $\sqrt{\langle n \rangle}$, with the field state remaining in the same coherent state $|\alpha\rangle = \sum_n c_n |n\rangle$. This is consistent with the semiclassical picture in which the field is unaffected by the atom, but the atom undergoes Rabi oscillation.

c) To capture more of the full picture, make a finer approximation, that

$$\sqrt{n+1} \approx \sqrt{\langle n\rangle + \delta n} \approx \sqrt{\langle n\rangle}\left(1+\frac{\delta n}{2\langle n\rangle}\right) = \sqrt{\langle n\rangle}\left(\frac{\langle n\rangle + n}{2\langle n\rangle}\right) \tag{28}$$

with $\delta n = n - \langle n\rangle$. Use this approximation to calculate the $|g\rangle$ and $|e\rangle$ contributions to the total wavefunction $|\psi_{e,n}(t)\rangle$; show that the state evolves approximately to

$$|\psi(t)\rangle \approx \frac{1}{\sqrt{2}}\left(|e\rangle|\alpha_+\rangle - |g\rangle|\alpha_-\rangle\right), \tag{29}$$

where

$$|\alpha_\pm\rangle = \frac{1}{\sqrt{2}}\left(e^{i\langle n\rangle\varphi(t)}|\alpha e^{i\varphi(t)}\rangle \pm e^{-i\langle n\rangle\varphi(t)}|\alpha e^{-i\varphi(t)}\rangle\right) \tag{30}$$

where

$$\varphi(t) = \frac{g_0 t}{2\sqrt{\langle n\rangle}}. \tag{31}$$

What is the state of the oscillator upon measuring the atom in state $|g\rangle$ or $|e\rangle$ at t such that $g_0 t/\sqrt{\langle n\rangle} = \pi$?

With $\varphi(t)$ as given, we can write

$$\begin{aligned} g_0\sqrt{n+1}t &\approx g_0 t\sqrt{\langle n\rangle}\left(\frac{\langle n\rangle + n}{2\langle n\rangle}\right) && (32)\\ &= g_0 t\frac{\sqrt{\langle n\rangle}}{2}\left(1+\frac{n}{\langle n\rangle}\right) = \varphi(t)\langle n\rangle + \varphi(t)n. && (33)\end{aligned}$$

Plugging this into our sum for the total state,

$$\begin{aligned} |\psi(t)\rangle &\approx \sum_n c_n\left(\cos(\omega_{n+1}t)|e,n\rangle - i\sin(\omega_{n+1}t)|g,n\rangle\right)\\ &= \sum_n e^{-|\alpha|^2/2}\frac{\alpha^n}{\sqrt{n!}}\left(\cos(\varphi(t)\langle n\rangle + \varphi(t)n)|e,n\rangle - i\sin(\varphi(t)\langle n\rangle + \varphi(t)n)|g,n\rangle\right)\\ &= \frac{1}{2}\left(|e\rangle\otimes\left(e^{i\varphi(t)\langle n\rangle}|\alpha e^{i\varphi(t)}\rangle + e^{-i\varphi(t)\langle n\rangle}|\alpha e^{-i\varphi(t)}\rangle\right) - |g\rangle\otimes\left(e^{i\varphi(t)\langle n\rangle}|\alpha e^{i\varphi(t)}\rangle - e^{-i\varphi(t)\langle n\rangle}|\alpha e^{-i\varphi(t)}\rangle\right)\right)\end{aligned}$$

Where the second-to-third step was simply expanding the cos and sin as complex exponentials. The last line is the result. At the time such that $\varphi(t) = \pi/2$ as stated which is the time $t = t_r/2$, with t_r the revival time, we have

$$|\psi(t_r/2\rangle = \frac{1}{2}\left[|e\rangle\left(e^{i\frac{\pi}{2}\langle n\rangle}|i\alpha\rangle + e^{-i\frac{\pi}{2}\langle n\rangle}|-i\alpha\rangle\right) + |g\rangle\left(e^{i\frac{\pi}{2}\langle n\rangle}|i\alpha\rangle - e^{-i\frac{\pi}{2}\langle n\rangle}|-i\alpha\rangle\right)\right]. \tag{34}$$

Upon measuring the atom in either $|e\rangle$ or $|g\rangle$,we see in either case we project the field state into a state that is an equal superposition of $|i\alpha\rangle$ and the opposite $|-i\alpha\rangle$, with a relative phase depending on the measured state of the atom. This is a superposition of opposite phases of the coherent state, and since for large $\langle n\rangle$ these would be macroscopically distinguishable states, these states are often called Schrödinger cat states of the field.

Question 3. Transient atom-photon entanglement during spontaneous emission

This exercise has some simple fun with full and reduced density matrices, using spontaneous emission as an example.

a) The (pure) state of a two-level atom and optical field during a spontaneous emission event was written in the notes effectively as:

$$|\psi(t)\rangle = a_e(t)|e,0\rangle + \sum_m a_m(t)|g,1_m\rangle; \tag{35}$$

where $a_e(0) = 1$ and in this notation m indexes the modes the atom can emit into, all initially in the vacuum state.

Taking $a_e(t) = e^{-\Gamma t/2}$ (as found under the approximations employed in the lecture), evaluate the trace of the total density matrix $\rho = |\psi(t)\rangle\langle\psi(t)|$ over all the possible field eigenstates to show that the reduced density matrix of the atom alone as a function of time during this process is

$$\hat{\rho}_A(t) = p_e(t)|e\rangle\langle e| + p_g(t)|g\rangle\langle g|, \tag{36}$$

with $p_e(t) = e^{-\Gamma t}$. $p_g(t)$ of course follows from normalization; relate it also to the coefficients $a_m(t)$.

The trace over field degrees of freedom F can be written as

$$\hat{\rho}_A(t) = \mathrm{Tr}_F(\hat{\rho}) = \langle 0|\hat{\rho}|0\rangle + \sum_m \langle 1|_m \hat{\rho}|1\rangle_m \tag{37}$$

$$= |a_e(t)|^2 |e\rangle\langle e| + \sum_m |a_m(t)|^2 |g\rangle\langle g|, \tag{38}$$

from which we see directly that $\hat{\rho}_A(t) = p_e(t)|e\rangle\langle e| + p_g(t)|g\rangle\langle g|$, with

$$p_e(t) = |a_e(t)|^2 = e^{-\Gamma t}, \tag{39}$$

$$p_g(t) = \sum_m |a_m(t)|^2 = 1 - e^{-\Gamma t} \tag{40}$$

where the last equality follows from $\mathrm{Tr}(\hat{\rho}) = 1$.

b) Entanglement between two subsystems within an overall pure state is reflected by the reduced density matrix of one subsystem (resulting from a trace over degrees of freedom corresponding to the other subsystem) describing a mixed state.

The Von Neumann entanglement entropy provides one quantification of this; calculate this quantity between the atom and the field, defined as

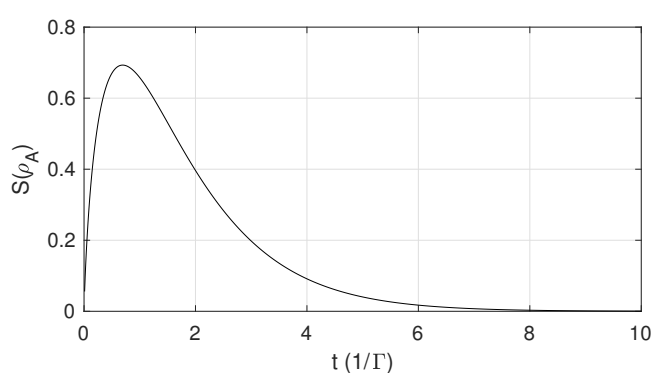
$$\mathcal{S}(\hat{\rho}_A(t)) \equiv -\mathrm{Tr}(\hat{\rho}_A(t)\log\hat{\rho}_A(t)). \tag{41}$$

Plot and comment on the evolution of $\mathcal{S}(\hat{\rho}_A(t)$.

Since ρ_A is diagonal, its log is a also a diagonal matrix with entries given by the log of the entries of ρ_A. Hence

$$\mathcal{S}(\hat{\rho}_A(t)) = \Gamma t e^{-\Gamma t} - (1 - e^{-\Gamma t})\log(1 - e^{-\Gamma t}). \tag{42}$$

This is plotted below: We see that the degree of entanglement briefly peaks, as the system exists



in a superposition of having emitted a photon and not (and the corresponding photon states) and then decays as the atom is increasingly likely to have decayed already.

c) What is the maximum $\mathcal{S}$ a two-level system entangled with other degrees of freedom can have? From your result in (b), either derive or argue at which t this entanglement entropy is maximized and the value it takes.

The maximum $\mathcal{S}$ would occur when the atom is in a 50/50 mixed state upon tracing over the field degrees of freedom, as with any two-level system. Hence the maximum value for a two-level system is $\mathcal{S}_{\max} = -\log(1/2) \approx 0.693$. Hence the atom is briefly maximally entangled with the field. This occurs when $e^{-\Gamma t} = 0.5$ or at $t = -\frac{1}{\Gamma}\log(\frac{1}{2})$.